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A Blockchain-Based Framework for COVID-19 Detection Using Stacking Ensemble of Pre-Trained Models

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ABSTRACT

In recent years, COVID-19 has impacted millions of individuals worldwide, resulting in numerous fatalities across several countries. While RT-PCR technology remains the most reliable method for detecting COVID-19, this approach is expensive and time-consuming. As a result, researchers have explored various machine learning and deep learning-based approaches to rapidly identify COVID-19 cases using X-ray images. Machine learning based models can reduce costs and have shorter processing times. However, preserving patient confidentiality poses challenges within such third-party-controlled systems, potentially failing to safeguard patients from potential disgrace and discomfort. Nonetheless, blockchain technology offers the potential to securely store sensitive medical data anonymously, without requiring third-party intervention. Consequently, the combination of deep learning and blockchain might offer a viable solution to mitigate the spread of COVID-19 while ensuring patient privacy protection. In this paper, we propose a hybrid model of blockchain and deep learning model for automatically detecting COVID-19 using chest X-rays (CXR). The deep learning model includes a stacking ensemble of three modified pre-trained Deep Learning (DL) models: VGG16, Xception, and DenseNet169. The model obtained an accuracy of 99.10% and 98.60% for binary and multi-class respectively. Further, to ensure COVID-19 patients' privacy and security, the Ethereum blockchain has been adopted to store information related to COVID-19 cases. In addition, a smart contract on the blockchain has been designed for handling X-ray images in the Interplanetary File System (IPFS).

1. Introduction

The outbreak of the coronavirus disease (COVID-19) initially emerged in December 2019 in Wuhan, China. However, this soon spread to other countries of the world [1,2]. In addition, this virus can be transmitted by hand-to-hand contact with a COVID-19-infected patient or through the air [3]. This virus, known as severe acute respiratory syndrome coronavirus-2, is the source of the COVID-19 epidemic disease (SARSCoV-2). Coronaviruses (CoV) are responsible for many illnesses, including SARS-CoV as well as MERS-CoV [4].

The most common variety today is the South African-discovered genotype Omicron-B.1.152a. [5]. Globally, 661,527,333 coronavirus cases have been reported since around December 25, 2022, with 6,685, 319 fatalities. A total of 634,147,227 persons have fully recovered from COVID-19 [6]. Seniors are more likely to get COVID-19 than younger people [7]. COVID-19 is the most prevalent among the elderly in the United States, India, and France [8]. Heart attacks, septic shock, pulmonary edema, and sudden renal failure are some of the severe complications caused by COVID-19 disease [9]. Therefore, quickly identifying and isolating COVID-19-infected patients is crucial for controlling and managing the pandemic [10,11].

COVID-19 symptoms range from fever, coughing, and shortness of breath to exhaustion, body aches, and headaches. Other signs include loss of taste or smell, sore throat, and congestion. Some may experience nausea, vomiting, or diarrhoea [12]. The virus primarily targets the lungs, leading to pneumonia and it can lower the oxygen level in the body. COVID-19's incubation period varies from two to fourteen days, with symptoms typically appearing within six to seven days after infection [13].

The present COVID-19 detection system is called real-time reverse transcription-polymerase chain reaction (RRT-PCR), which requires an individual's nasopharyngeal swab. However, it takes a day or more for the test results to be available. In the meantime, the infected patient might spread the virus. Therefore, there needs to be a quick and efficient

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Table 1
List of acronyms.

CXR	Chest X-Ray
DL	Deep Learning
ML	Machine Learning
IPFS	Interplanetary File System
BC	Blockchain
CoV	Corona Viruses
RRT-PCR	Real-time Reverse Transcription-Polymerase Chain Reaction
MRI	Magnetic Resonance Imaging
CAD	Computer-Aided Diagnostic
CNN	Convolutional Neural Network
CT-scan	Computed Tomography scan
GAP	Global Average Pooling
TL	Transfer Learing
PoW	Proof of Work
PoS	Proof of Stake
DPoS	Delegated Proof of Stake

testing approach. Recently, many researchers have explored various machine and deep learning models to detect COVID-19 cases. However, findings have many false negatives due to the low sensitivity of COVID-19 detection, which might ultimately increase the disease's transmission [14-16].

Training machine learning and deep learning algorithms using CXR is a highly effective method for diagnosing pneumonia and COVID-19. Compared to computed tomography (CT) and magnetic resonance imaging (MRI), chest X-rays expose patients to a lower radiation dosage. A CXR is much harder to diagnose manually than other imaging modalities like CT or MRI [17]. The CXR analysis is a time-consuming, labor-intensive, and lengthy process that should be automated to save time and effort [18].

Computer-aided diagnostic (CAD) technology, using automated deep learning systems, has shown promising results in accurately classifying medical data, particularly in CXR (chest X-ray) images [19]. Physicians widely use CAD systems to swiftly and precisely evaluate and interpret medical imaging data [20].

However, the challenge in using machine learning or deep learning to detect COVID-19 cases lies in securely storing and tracking positive cases in a decentralized manner without compromising patients' privacy. In addition, the traditional COVID-19 tracking approach is centralized and doesn't adequately protect patient privacy, leading to affected individuals feeling socially embarrassed about their condition.

Blockchain technology has recently appeared as one of the most secure storage systems to preserve patients' sensitive data. Blockchain technology's decentralization, transparency, and immutability features have made it popular among researchers. However, blockchain is not appropriate for big data such as medical images and IoT data due to the requirement of the high volume of storage. Therefore, many researchers have proposed the InterPlanetary File System (IPFS), a peer-to-peer network like blockchain, to store medical images to tackle this issue. A user interface can interact with the IPFS using Web3 programming to store medical images. Integrating the deep learning model with blockchain technology can solve the privacy and security issues of COVID-19 patients. To address the privacy issues mentioned above, we proposed a blockchain-based COVID-19 detection framework that includes a smart contract system to upload COVID-19-positive case-related information in the blockchain. The acronyms that are used in this paper are provided in Table 1:

Our contributions are summarized as follows.

- To perform rigorous pre-processing on the COVID-19 dataset involving normalization, image rescaling, image enhancement orientation, and data augmentation, applying ens-net to remove text from the images.
- To propose a stacking ensemble based deep learning approach which consists of modified pre-trained models, including VGG-16, Xception, and DenseNet169, to detect COVID-19 cases where hyper-tuning is applied to increase the model's accuracy.
- To develop a smart contract-based COVID-19 positive cases storage framework. The smart contract is de-ployed on the Ethereum blockchain. The smart contract enables the storage of hash code as well as sensitive data of the patient in the IPFS and blockchain.

The rest of the paper is organized as follows. In section 2, we provide literature retated to our work. In section 3, the data sets and proposed model are described . Section 4 presents the details of the experiments and findings of this paper. Finally, Section 5 concludes with possible future work.

2. Literature Review

Detection of COVID-19 using various clinical imaging techniques is a rapidly expanding area of research. Not many researchers use machine learning-based techniques to detect COVID-19 disease, but many studies have applied deep learning-based approaches to correctly identify COVID-19 cases where convolutional neural networks are most used for successfully recognizing COVID-19 from X-ray images. Recent studies showed that chest X-rays can assist in diagnosing COVID-19 promptly and contain the spread of COVID-19 further.

Lafraxo et al. [21] suggested a deep learning model named CoviNet, in order to detect COVID-19 using CXR pictures. Suggested structure relies on the pre-processing steps of a CNN, histogram equalization, and an adaptive median filter. The model was fully trained using a publicly accessible dataset. CoviNet features two phases: an automatic extraction of features phase and a classification phase. The proposed CNN contains four convolutional layers, where the filter size is 3×3, and 32 filters total across the three convolutional layers. In the fourth layer, 64 filters are applied. The activation function "Relu" is applied after each convolution layer. The model's performance was assessed and retrained varying architectures and tuning hyperparameters until the highest accuracy score was obtained. The model's accuracy for binary classification and multi-class (COVID-19, normal, Pneumonia) classification were 98.62% and 95.77%, respectively.

Ahmed et al. [22] proposed a model where deep CNN is used for the detection of Covid-19 using chest X-rays images. Two Kaggle datasets were used to gather X-ray images. Five convolutional layers make up this proposed model, and each is followed by dropout, batch normalization, and max pooling layers. A completely linked layer with 512 neurons processes the final convolutional layer, and the final layer has three neurons for each type of X-ray. The activation function for every layer was ReLU, while the final dense layer utilized softmax. By implementing this deep learning approach, an accuracy of 90.64% and an F1-Score of 89.8% were attained.

Rahman et al. [23] proposed a model on how various image-enhancing methods affect deep CNNs' ability for automated detecting of COVID-19 from CXR pictures. In addition, unique U-Net architectural variant was suggested that beat the most recent U-Net model for lung segmentation from CXRs. Here, amodel using CNN was built from scratch and imageNet weights were used to train six different DL-CNN models. Seven CNN models' results for 5 distinct image enhancement techniques were used to analyze the categorization. They achieved an accuracy of 96.29% without segmentation and 95.11%, with segmentation, respectively. DenseNet201 fared better for segmented lungs with a gamma enhancement approach, but the ChexNet model having gamma enhancement approach performed best without picture segmentation.

Loey et al. [24] proposed a novel Bayesian optimization-based CNN model, for the recognition of chest radiograph images (CXRs). There are two primary parts to the suggested model. The first one extracts and learns deep features using CNN. The second one is a Bayesian image optimizer, which adjusts CNN hyper-parameters according to the line with the target function. 10,848 photos, including 3616 images of each class, make up this huge, balanced dataset that was used. Bayesian optimization is compared to three different scenarios. In the initial scenario, we found that the 96% accuracy of the Bayesian search yielded optimal architecture.

Aggarwal et al. [25] suggested a standard DL framework to accelerate the differentiation between COVID-19 patients and other groups with pneumonia, a comparative comparison of several deep learning architectures has been conducted. The models included in this investigation are InceptionV3, NASNetMobile, MobileNetV2, ResNet50, VGG16, Xception, InceptionResNetV2, and DenseNet121. Two datasets were analyzed for this research project. Three groups of photos are included in Dataset-1: COVID-19, Normal, and Pneumonia. In contrast, 2nd dataset includes similar classes but has a stronger emphasis on the two main types of pneumonia: viral and bacterial pneumonia. The accuracy for the first dataset was 97% using DenseNet121, and the accuracy for the other dataset was 81% for MobileNetV2, which provided the best output.

Lasker et al. [26] suggested LWSNet, which is a lightweight stack design for differentiating Covid-19 from other groups with pneumonia and non Covid-19. Modified Inception, ResNetV2, CLCNN, MobileNetV2 and VGG19 architecture were used for the classification. On these four custom models, the stack ensemble method was also used. Double, triple, and quadruple stacking were three different stacking methods that were tested. Using datasets 1, 2, 3, and all together, the stacking ensemble approach that utilized quadruple stacking achieved the highest accuracy rates, which were 97.28%, 96.50%, 97.41%, and 98.54%, respectively.

Sahin et al. [27] suggested a novel CNN model that can automatically identify COVID-19 using CXRs. They compared the suggested model with models pre-trained, MobileNetv2 and ResNet50. The model was trained and tested on the prepared COVID-19 dataset (13,824 X-ray images). The model achieved an accuracy of 96.71%, with F-1 score of 91.89%.

Ramadhan et al. [28] proposed an approach that uses image cropping techniques and an updated VGG16 model using three databases. The model consists of 4 crucial stages: separating the data as training and test sets; cropping and resizing images as part of preprocessing; constructing a new VGG16-CNN model with three different classifications and last, a description of the model's assessment. For multiple classifications, the suggested model had an accuracy of 97.50%, while for binary classification, it had an accuracy of 99.76%. The VGG16 model's parameter count has been reduced from 138 million to 40 million.

Heidari et al. [29] discussed a model that receives minimal data and trains DL models by using them in a blockchain-based CNN method. In order to avoid the necessity for layers to be initialized randomly, transfer learning was applied. The data is verified via the Blockchain system, and the model is trained globally using the DL approach while maintaining the institution's privacy. The BDLCD technique refers to a COVID-19 screening and diagnosis system for chest CT images that are based on blockchain, utilizes DL, and focuses on edge computing. Five different datasets were used, and the achieved results for precision, recall, F1, and accuracy were 2.7%, 3.1%, 2.9%, and 2.8%, respectively.

Dar et al. [30] suggested a model for data integrity, management, and real-time data handling. The management of COVID-19 data using blockchain technology is demonstrated in this study. The data collection layer, the data privacy and access layer, as well as the data information storage layer, are the three fundamental framework levels that are investigated. Data privacy and access layer: it regulates who or how can have access to the information, making it a contributing element of the framework. It gathers information for the data collection layer from cellphones, medical equipment, and other information-compiling tools. Data access layers are implemented via smart contracts. Additionally, the patient's information is stored in databases that are decentralized and made available by various hospitals and remote users on the data storage and analysis layer.

Table 2
Comparison of related works of COVID-19 diagnosis process using CXRs.

Author's	Number of class	Techniques used	Accuracy (%)
Lafraxo et al. [21]	2	CoviNet	98.62
	3		95.77
Ahmed et al. [22]	3	CNN	90.64
Rahman et al. [23]	3	U-Net	95.11
Loey et al. [24]	3	CNN	96.00
Aggarwal et al. [25]	3	InceptionV3 + NASNet- Mobile+ MobileNetV2+ ResNet50 + VGG16 + Xception+InceptionResNetV2 + DenseNet121	97.00
Lasker et al. [26]	3	LWSNet	96.50
Sahin et al. [27]	2	CNN	96.71
Ramadhan et al. [28]	2	VGG-16	97
	3		95.14

Song J et al. [31] suggested a new system for sharing COVID-19 information and alerting people to risks on a global scale. This system is built on a combination of Bluetooth technology, smart contracts, as well as the Blockchain. In order to protect consumer privacy, a unique platform that uses the Blockchain to communicate accurate and unmodified contact monitoring data from various sources has been created. The proposed Covid-19 data sharing and risk alerting system collect data from two facets: indirect contact tracing using location and direct contact tracing using Bluetooth. The system is composed of four levels: the file storage layer, the smart contract service layer, the cellphone service layer, and the consumer interaction layer. Additionally, it provides two key trace and notification services: a contact tracing service based on location and a personal contact tracing service based on Bluetooth.

A blockchain framework for the vaccine distribution chain was developed by Kamenivskyy et al. [32]. This framework can reduce the circulation of fake vaccines and vaccination records, enhance stakeholder communication, increase data security, decrease the likelihood of a successful cyberattack, streamline the handling of vaccines in the provider's facilities, and help providers digitize their facilities to prevent storage issues. The user, blockchain, application, network, and data layers are the six levels that are employed.

Kumar et al. [33] suggested a framework that may use a minimal number of computed tomography (CT) images from various sources, train a global DL model that uses federated learning based on blockchain, and share the information among hospitals while maintaining privacy. Additionally, COVID-19 patients are detected utilizing segmentation and classification based on Capsule Networks. The suggested model consists of a locally used model and federated learning based on blockchain. Issues of diverse CT-scan data are addressed initially, which is dealt with using the data normalization technique. Then, the local model is trained to recognize COVID-19 patterns after segmentation using SegCaps. Finally, the weights of the local model across the blockchain network are distributed for training the global model.

A privacy-preserving contact tracing technique, BeepTrace, introduced by Xu et al. [34], uses blockchain tech-nology to connect users and authorized solvers while desensitizing user ID and location data. With the added benefits of being battery-friendly and available worldwide, this technique demonstrates improved security and pri-vacy. To facilitate the analysis of geodata and passive notification, BeepTrace suggests an architectural view of blockchain address and transaction design utilizing two chains. By utilizing two distributed blockchains, a novel method is provided to decouple user privacy. Authorized solvers can access the tracing chain with desensitized personal location

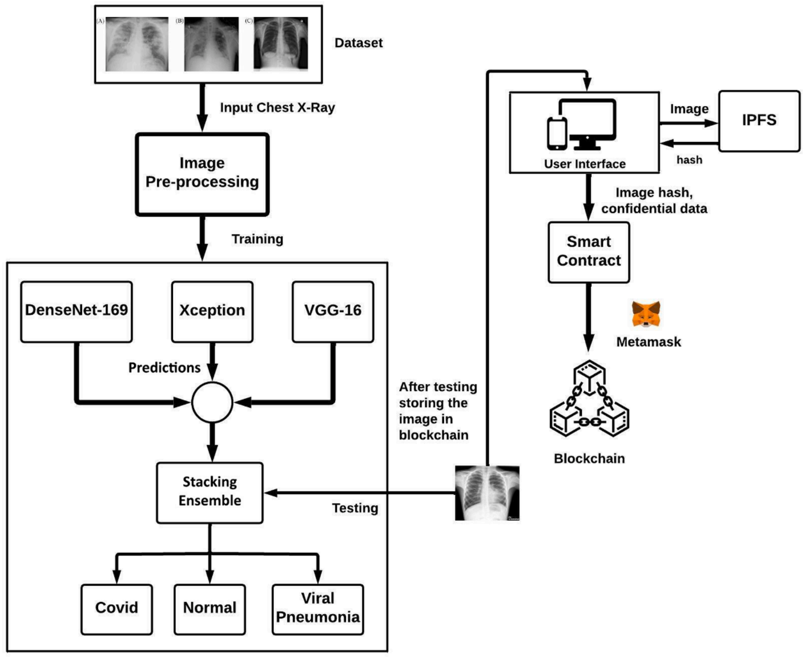

Fig. 1. Flow diagram of the proposed system.

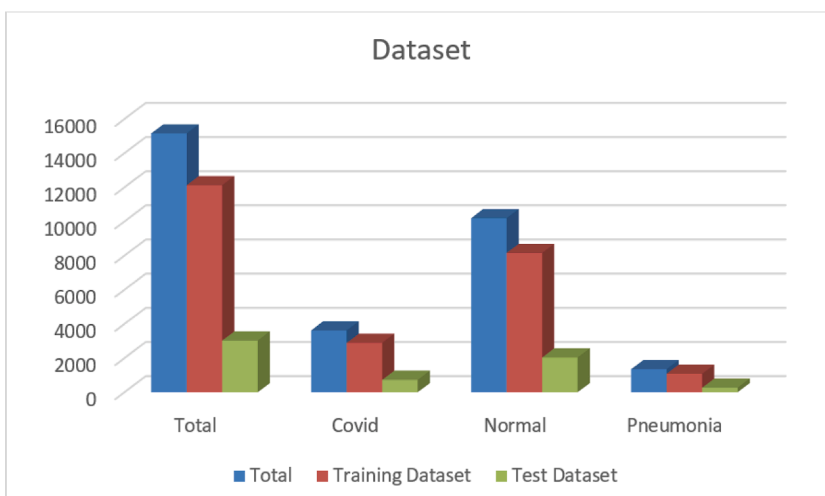

Fig. 2. Categorization level of dataset.

information for contact matching. Also, there is a notification chain where the local self-matching results for exposed users (using only a pseudonym or its fingerprint) are released.

As shown in (Table 2.), numerous studies have attempted to detect COVID-19 infections from CXRs using various DL techniques [19–26]. Further investigation is necessary to address several shortcomings in the COVID-19 detection and diagnosis process that depend on CXRs, according to the analysis to get an insight into the existing key technologies and research gaps.

3. Methodology

In this section, the suggested methodology and architecture are explained and after implementing these the performance analysis was conducted. Fig. 1 provides a broad overview of the proposed methodology. Proposed COVID-19 detection model includes data collection, image preprocessing, model training, classification (binary, multiclass), and storing the data in the blockchain.

3.1. Data Acquisition

To train our model, we collected chest X-ray images from an open-source dataset [35]. This dataset includes information from patients with COVID-19, pneumonia (induced by illnesses including MERS, SARS, and ARDS), and information on normal X-ray. This dataset were collected from different hospitals and clinical trials. The dataset contains four different categories: COVID-19, lung opacity, normal, and viral pneumonia images. The dataset has primarily 219 COVID-19 photos, 1341 images of normal cases, and 1345 images of viral pneumonia. After modifying the dataset, there are now 3,616 images of COVID cases, 10, 192 images of usual cases, and 1,345 images of pneumonia, as well as 6012 lung opacity image cases. We removed the lung opacity images from the dataset for this research. Fig. 2 depicts the categorization of the dataset.

Table 3
Training and testing details of the dataset.

	Total	Covid	Normal	Pneumonia
Total	15153	3616	10192	1345
Training Dataset (80%)	12122	2892	8154	1076
Test Dataset (20%)	3031	724	2038	269

The study employed two classification methods, namely binary classification (normal cases vs COVID-19 cases) also multi-classification (3 classes: normal cases, COVID-19, and pneumonia). The dataset was divided into two segments, with 80% used for training and 20% for testing. Table 3 displays the quantity of images utilized from this dataset.

We applied several pre-processing methods on the dataset. This pre-processing involves normalization, image rescaling, image enhancement, and data augmentation.

- All the images in the dataset are resized to the same 224*224 pixel for training and testing purposes.
- Next, outliers and other unnecessary photos are eliminated. We perform various image orientations such as scaling, cropping, rotating, sheering, horizontal flip-flopping and zooming.
- In order to properly train deep learning models, image datasets are expanded using image augmentation techniques. Besides, overfitting must be avoided by using data augmentation. We use Albumentations9 library for augmenting images because of its large collection of augmentation functions, which includes course dropout, elastic transform and different picture distortions.
- Finally, EnsNet is used to remove any text from the images.

After pre-processing, we extract features and use a deep learning model to classify the images more accurately.

3.2. Model Training and Classification

Deep learning techniques are an effective approach for classifying images from the healthcare industry. Basically, it has been employed to detect and classify different diseases. The proposed model is based on a stacking ensemble of modified DenseNet169, Xception, and VGG16 architectures. For these methods, more information is provided in the sections that follow:

3.2.1. DenseNet169 Architecture

This model is composed of two parts: a feature extraction part as well as a classification part. The modified DenseNet169 architecture that is used as an input layer is 224 by 224 and has three channels. This is followed by a conv2d layer, and after that, it passes through the conventional DenseNet169 model. A GAP layer is used, followed by a 1664-channel batch normalization layer. After that, the dense layer takes 512 channels as inputs, and the output is 128 channels. This batch normalization and dense layer repeat three times with an output of channel size 32. Then, two layers of batch normalization occur, where the input and output channels are each 32, followed by the last dense layer, where the output is of three channels. In Fig. 3, the DenseNet169 architecture of our method is shown:

This model consists of three blocks of layers that are fully connected. The first block consists of Global Average Pooling with Batch Normalization and a dense layer. The next block consists of two layers of Batch Normalization with a dense layer afterwards. Finally, the last block is combined with two batch normalization and one dense layer. The first two layers used the ReLU activation function, and the last one used the Softmax activation function.

3.2.2. Xception Architecture

The input layer is 224 by 224 and has three channels. This is followed by a conv2d layer, and after that, it passes through the conventional

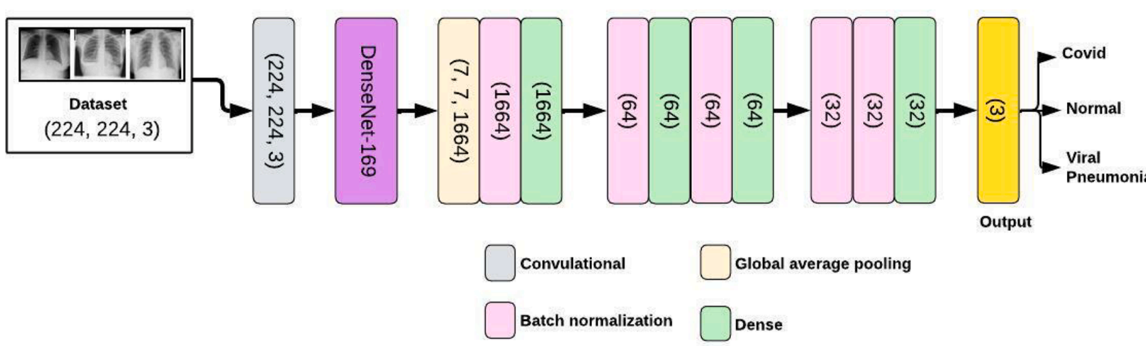

Fig. 3. Modified DenseNet169 Architecture.

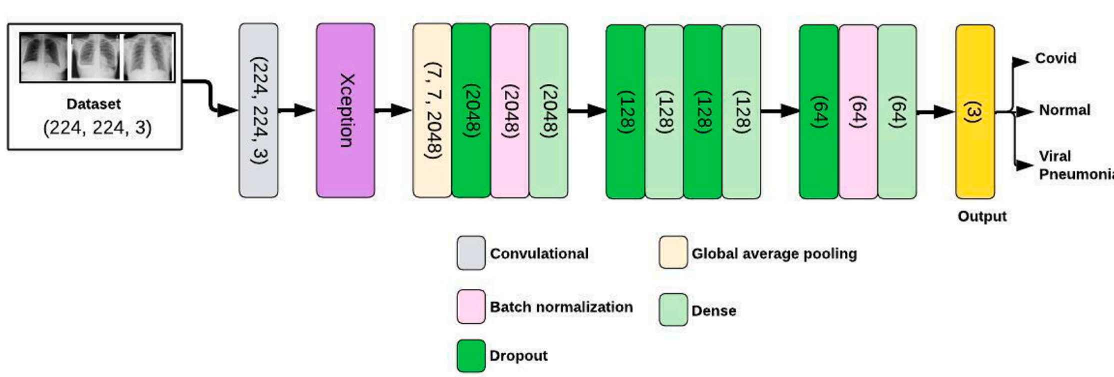

Fig. 4. Modified Xception Architecture.

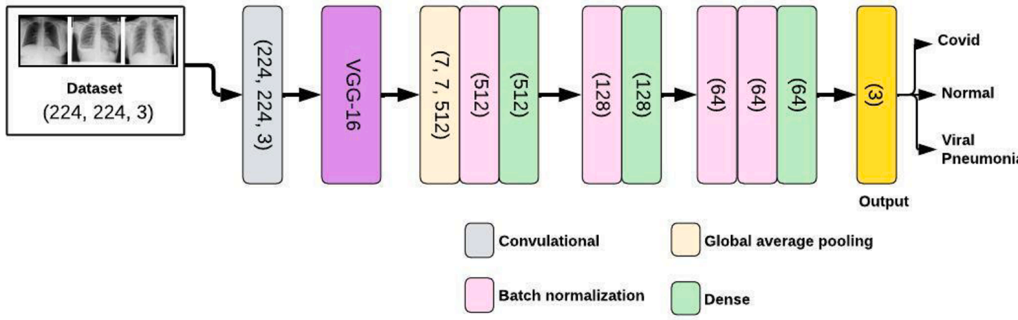

Fig. 5. Modified VGG-16 Architecture.

Table 4
Augmentation method values.

Methods	Values
Rescale	1/255
Shear range	0.2
Width Shift range	0.2
Height Shift range	0.2
Random Zoom range	0.2

Table 5
Hyper-parameter tuning values.

Hyper-parameter tuning	values
DL Framework	tensorflow
Number of epochs	125
Batch size	32
Optimizer	Adam
Learning rate	1e-3
Loss function	Cross Entropy Loss
Random Horizontal Flip probability	True
Random Vertical Flip probability	True
Random Resized Crop size	224
Random Rotation angle range	360 degree
Random Resized Crop Scale	0.5

Xception model. A GAP layer is used, followed by a dropout layer and a batch normalization layer. Next, three layers of dense and dropout layers pass the following image, which goes through batch normalization. The first dense layer takes 2048 channels as inputs, and the last dropout layer has an output of 64 channels. Then, the last batch normalization occurs, where the input and output channels are each 64, followed by the last dense layer, where the output is of 3 channels. In Fig. 4, the Xception architecture of our method is shown in the details:

This model is similar to Densenet-169, but it uses pre-trained Xception instead of pre-trained Dense-169.To avoid over-fitting the model uses training data, also, a dropout layer is inserted between the layers that are fully connected, with a probability of 0.5.

3.2.3. VGG16 Architecture

In order to create a COVID-19 detection model using CXRs, we modified the VGG16 CNN model. The input layer is 224 by 224 and has 3 channels. This is followed by a conv2d layer and it passes through the convolutional VGG16 model. A GAP layer is used, followed by a 512-channel batch normalization layer. After that, the dense layer takes 512 channels as inputs, and the output is 128 channels. This batch normalization and dense layer repeat with the output of channel size 64. Then, two layers of batch normalization occur, where the input and output channels are each 64, followed by the last dense layer, where the output is of three channels. In Fig. 5, the VGG16 architecture of our method is shown in details:

This part consists of pre-trained VGG-16 and three extra blocks of fully connected layers. The first block consists of Global Average Pooling with Batch Normalization and a dense layer. The next block consists of one layer of Batch Normalization with a dense layer. The last block is combined with two batch normalization and one dense layer. The ReLU activation function is used in the first two fully connected layers, and the last one uses the Softmax activation function.

3.2.3.1. Data augmentation and Hyper-parameter tuning. Overfitting is one of the major issue with image datasets while training a DL model. To overcome this issue, images in the dataset are increased and generalized using augmentation. Various augmentation techniques have been used in this work, including horizontal flipping, range shearing, range zooming, and shifting.

After analyzing the validation set, the optimal values for the hyper-parameters - Random Resized Crop scale, Random Rotation angle range, Random Resized Crop size, and Random Horizontal Flip probability - have been identified. To tune hyperparameters, the grid search method has been used.

3.2.3.2. Training details. Table 4 and Table 5 display the training specifics and variables that were consistent throughout the trial:

3.3. Stacking ensemble and classification

Stacking ensemble model [36] is a technique that combines multiple models to improve the performance of the final model. In case of deep learning, a stacking ensemble model is a combination of multiple deep learning models, each trained on the same dataset. The predictions from each individual model are then combined to make the final prediction. Consequently, a CNN model with a stacked ensemble will be highly accurate and generalised.

The predicted class is then obtained by passing the outputs of each of the three components via a single neuron, as shown in Fig. 6. This single neuron employs the argmax activation function. The stacking model comprises of one neuron, which allocates weights to the outputs generated by the three distinct sections. Subsequently, it uses these weights and also outputs from the 3 different sections to forecast the output class, including COVID-19 positive, negative, or pneumonia. To train the suggested model rapidly, transfer learning is used.

3.4. Proposed Blockchain Model

In this section, we describe a decentralized application which is developed to preserve COVID-19 cases related information. The application includes a User Interface, a deep learning model, IPFS, a smart contract and an Ethereum blockchain. The smart contract aids in tracking COVID-19 cases and implements access control for relevant documents stored in the blockchain. The blockchain is not appropriate technology to store large amounts of documents due to the requirement of a higher amount of memory. Therefore, we use Infura platform which facilitates IPFS based storage for storing COVID-19 cases related

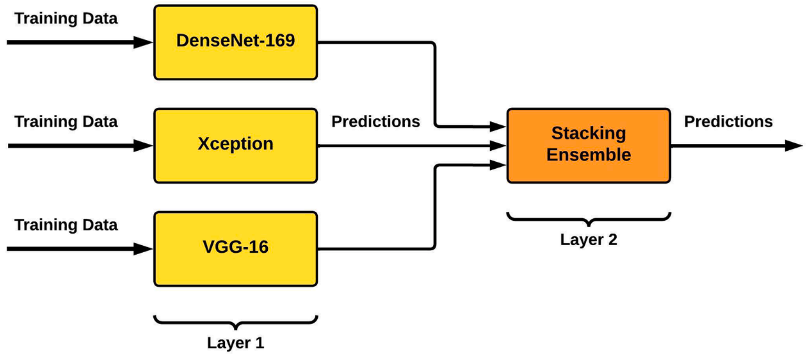

Fig. 6. Stacking ensemble used in this research work.

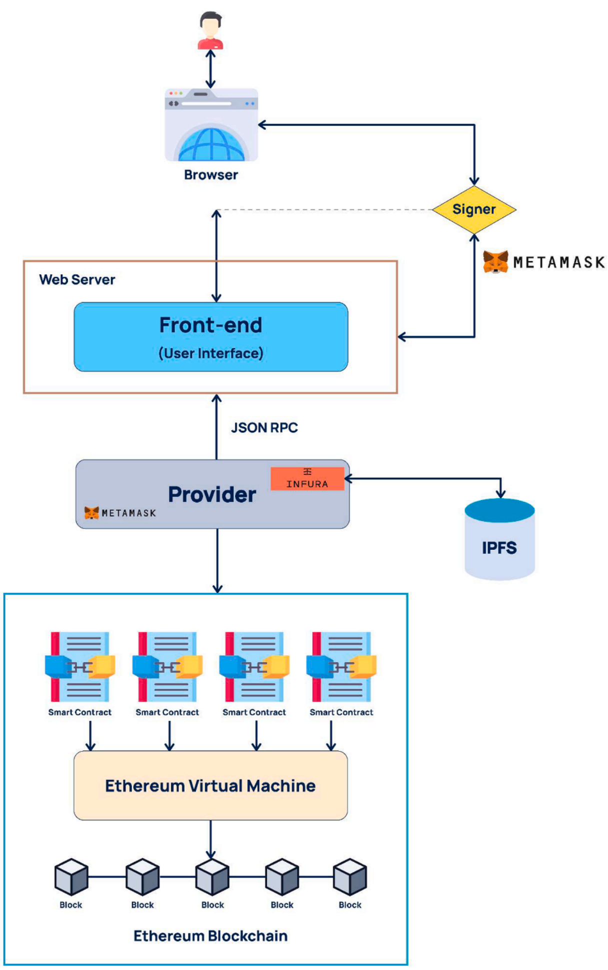

Fig. 7. Proposed Blockchain Model.

information. Since IPFS provides decentralized storage, there is no single point of failure and bottleneck issue.

Fig. 7 shows the overall architecture of the proposed decentralized application. First, a deep learning model is trained to detect COVID-19 cases correctly. Secondly, the deep learning model is connected with an User Interface. The user uploads a testing X-ray image through UI and then send it to the trained deep learning model. The deep learning model returns outcome to the UI. Next, the UI sends the image to IPFS which returns the hash code of the image to the UI. To ensure integrity of data saved in IPFS, the hash code is stored in the blockchain, which enables users to validate data. Finally, the UI makes a transaction that contains hash code of the tested X-ray image, and patient basic information. A smart contract" is created and deployed in the blockchain to process the transaction in the BC. The smart contract directly and automatically regulates the transfer of secured data between different stakeholders.

3.4.1. Potential keys of blockchain

One of the key components of the proposed method is the blockchain. Several key aspects of blockchain make it a promising foundation for the proposed method, including:

- Immutability: Storing information or transactions on a blockchain makes it incredibly challenging, if not almost impossible, to modify, disrupt, or erase them. The immutability of blockchain is facilitated by cryptographic hashing, which generates a unique digital fingerprint, or hash, for every data block. Even a small alteration in the data will result in an entirely different hash. This characteristic of immutability in blockchain ensures a significant level of transparency and integrity.
- Decentralization: In a blockchain network, there is no central authority or a single entity that has complete control over the system. Instead, a group of nodes collectively maintains the network, decentralizing it by participating in the validation and verification of transactions and storing copies of the blockchain ledger. This decentralized approach enhances security by eliminating dependence on a single point of failure, as the network can still operate even if some nodes experience issues. Moreover, the involvement of multiple independent parties in verifying and recording transactions and data enhances transparency and reduces the risk of censorship or manipulation.
- Security: Cryptographic hashes are employed to encrypt all data elements within the blockchain, ensuring their security and integrity. This cryptographic mechanism removes the need for a centralized authority, as it prevents any individual from altering network properties for personal benefit or gain.
- Consensus: Consensus is a vital aspect of maintaining the integrity and security of a system, enabling multiple untrusted nodes to reach agreement on a specific data block. Various consensus methods, such as Proof of Work (PoW), Proof of Stake (PoS), and Delegated Proof of Stake (DPoS), exist to facilitate this process.
- Transparency: Blockchain facilitates transparency by enabling unrestricted access to transactions and stored data. This transparency, coupled with the immutability of documents, enhances auditability in various domains such as supply chain management and financial services. The transparency inherent in blockchain systems promotes accountability, instills trust among participants, and reduces the risk of fraudulent activities or manipulations.

3.4.2. Private and public blockchain

Private and public blockchains are two different types of popular blockchain networks which have distinct purposes and contrasting aspects. Key features of them include-

- Private Blockchain: A private blockchain, also known as a permissioned blockchain, is a restricted and centralized blockchain network. In a private blockchain, only a predetermined group of users or organizations are permitted to connect to the network and take part in the consensus process.
- Public Blockchain: A public blockchain, sometimes referred to as a permissionless blockchain, is a de-centralized system that enables anybody to take part in, view, and validate transactions. It is accessible to all users and runs on a trustless model where transactions are verified and the network's integrity is upheld through consensus algorithms such as Proof of Work (PoW) or Proof of Stake (PoS). A few examples of public blockchains include Bitcoin and Ethereum.

Our approach uses the public blockchain because it needs to be extremely secure against manipulation, all transactions are transparent, and anonymity is maintained because transactions spread across the public ledger as bits of data and cannot be traced back to the users' original addresses. We utilized the private Ethereum on our computer to upload our smart contract there and implement the decentralized application.

3.4.3. Blockchain platform and networks

Blockchain networks and platforms are decentralized systems that use blockchain technology to support safe and open data management and transaction processes. Bitcoin, Ethereum, Binance Smart Chain, Ripple, and Hyperledger Fabric are some examples of well-known blockchain networks and platforms.

The blockchain platform used in this paper is Ethereum. It enables the development of decentralized applications (DApps) and the execution of smart contracts. To keep track of all transactions and smart contract executions, Ethereum uses the blockchain, a distributed and decentralized ledger. Smart contracts are self-executing contracts with predetermined terms and circumstances. Smart contracts are carried out on the Ethereum network through the Ethereum Virtual Machine, which serves as a runtime environment. To carry out transactions, a fuel or "gas" called ether (ETH) is used by the network. Additionally, the Ethereum community switched from a proof of work (PoW) consensus process to a proof of stake (PoS) consensus mechanism, which uses less energy.

The term "on-chain transaction" refers to a transaction that is completed entirely on a blockchain network. The transaction is entered into a blockchain network's public ledger after it has been verified. "Off-chain transactions", in contrast, transfer part of the work from a blockchain system, which can then be reintegrated into a blockchain. This is needed when someone wants to check information using the blockchain but doesn't want to share the information publicly. Also, the information being stored might be too big for the blockchain to handle. Users of an off-chain network consent to a third party approving and authenticating transactions. Scalability, lower transaction costs, and increased privacy are a few advantages that off-chain solutions can offer. For that reason, we employed off-chain blockchain transactions.

A decentralized application is developed to store patients' confidential data and keep track of all transactions. With meta mask transaction confirmation, people can do tasks in the blockchain. For scalability purposes, only the data's hash is recorded in the blockchain; the data is kept in IPFS (a distributed file system and storage platform).

Blockchain cannot store files of large size; that's why files are saved in IPFS. After files are stored in IPFS, a hash is generated, then the hash is stored in the blockchain. There is no single point of failure since it is decentralized. A traceable and tracking system for medical products based on IPFS, blockchain, and deep learning is proposed, as shown in Fig. 7.

A software program known as a "Smart Contract" is created, which, under specific circumstances, directly and automatically regulates the transfer of secured data between the parties. The process starts by identifying CXRs and categorizing them as normal cases, covid-19 positive cases, or pneumonia affected using the DL methods. The image data are preprocessed before being stored in IPFS.

- Image Stored into IPFS: IPFS or Ethereum platform data storage paradigm is a free and open system for data encapsulation and data analysis. The peer-to-peer (P2P) distributed storage format offered by IPFS makes it simple to store massive amounts of medical records. IPFS uses a distributed hash table (DHT) to store files with their content-addressed hash and removes duplicate files using version control history. Additionally, IPFS locally generates a hash of files with high request frequency to guarantee speedy access the following time. The submitted images are largely collected and encapsulated by the data encapsulation module. The data management system module contains the IPFS, database, and blockchain. The data management system searches the database for the event hash of the medical photos on the blockchain and the operation content after acquiring the encapsulated data (IPFS hash address of the tracing data). To store an image in IPFS, we must first launch a daemon function. Obtaining the prior provenance data packet comes next. The new information is saved in the origination packet in order to create a new authenticity packet. When extracting content, the application uses the blockchain transactional hash to query transactions, receives the IPFS hash ip then gets the data using IPFS. Before sending the data back to the application layer for development, the data analysis module analyzes the information it receives from the data monitoring system. As we employed model storage off-chain IPFS, this solution can be utilized for tasks like- supply chain, auditing, storing patient diagnostic reports etc. We already know that in off-chain storage, A report is stored in IPFS, and in reply to that, it receives the content-addressed hash. This hash can be used in future to extract this report from the IPFS network.
- Create Hash Value: IPFS uses hashes to route files, allowing data to be shared globally without fear of loss or modification. The encrypted data to be shared are kept in IPFS, and only the hashes which is obtained from IPFS are transferred on the blockchain, lowering the data storage burden. To create an image hash value for images, a smart contract is implemented by Remix IDE. When it is deployed, then a request is sent to Etherscan through Metamask. It created a decentralized web that awaits and unlocks the metamask. Furthermore, an IPFS API called Infura was used to upload files to IPFS and then download them. When the applicant clicks Request, it automatically creates a new block. This hash is used when an image is searched and needs to be downloaded.

4. Results and Evaluation

The suggested model was implemented using Keras and Tensor Flow's GPU support. The model uses Adam optimizer, which uses stochastic gradient descent and adaptive estimation of first-and second-order moments. When tackling problems requiring a large number of datasets or components, this approach is quite effective. The optimization approach used a constant learning rate throughout training for modifications of all values. Adam adjusted its weights while working out using both the RMSProp and the AdaGrad technique. Adam is now one of the most widely used optimization algorithms, in large part because it offers both the smart learning rate annealing and momentum features of the algorithms that have been examined so far. In the proposed method, learning rate of 0.001 was employed, the value of beta 1 is set to 0.9, beta 2 to 0.999, epsilon to 0.1, and decay to 0. The proposed model was used to identify CXRs of three classes (covid, non-covid, and pneumonia). The following four metrics are used to assess the proposed architecture's performance outcomes:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Table 6
The statistical results of class based performance of proposed Densenet-169 model.

Class	Task	Precision	Recall	F1-score	Loss	Accuracy
Three class	Covid-19	0.96	0.99	0.97	0.077	0.974
	Pneumonia	0.99	0.97	0.98		
	Normal	0.97	0.97	0.97		
Two Class	Covid-19	0.99	0.97	0.98	0.051	0.981
	Normal	0.97	0.99	0.98		

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 * TP}{2 * TP + FP + FN}$$

4.1. Results of the Proposed System

The CXR image dataset was classified as binary or multi-class. Deep learning approaches have demonstrated excellent efficacy in diagnosing COVID-19. The proposed model employs a stacking ensemble model composed of the Densenet-169, Xception, and VGG-16 algorithms. For these methods, the statistical and visual findings were acquired separately, as shown below. Deep learning algorithms have been used in many research for model training and testing, as detailed in the literature review. In this work, the stacking ensemble method is proposed, which can identify a small characteristic and is thought to be an effective strategy.

4.1.1. Performance of DenseNet-169 Algorithm

Using the given dataset, the Densenet-169 technique is trained using parameters that perform much better than the default parameter configuration. For instance, 125 epochs produce better outcomes. Furthermore, with a loss value of just 0.07 and an accuracy of 97.77%, the suggested model utilizing the DenseNet-169 algorithm worked well. Table 6 provides the class-based performance results of DenseNet-169 model.

Table 6 displays precision, recall, F1-score, loss values and accuracy of the model after training and testing. The results demonstrate the effectiveness of the suggested methodology in identifying COVID-19. Fig. 8, provides an overview of the outcomes determined using the data and the epochs.

The model accuracy curve reveals that the training accuracy is high, but the initial performance of the model on the testing dataset is not satisfactory. The curve exhibits significant fluctuations. However, after 50 epochs, the curve shows an improvement in performance for both the training and testing datasets. The confusion matrix for the Densenet-169 algorithm with the provided dataset is presented in Fig. 9:

These parameters are necessary for deep learning algorithms to compute the loss value and accuracy while model training using a training dataset. These numbers might be positive or negative since it evaluates how well the model matches the training set of data.

4.1.2. Performance of Xception Algorithm

Another deep learning algorithm with updated characteristics, called Xception, was also utilized to identify and find COVID-19 and viral pneumonia diseases. For binary classification between COVID-19 and normal same dataset was utilized. As a multi-class classifier, the same dataset was again utilized for identifying COVID-19, normal cases, and viral pneumonia. For instance, 125 epochs produce better outcomes with a loss value of just 0.04 and an accuracy of 96.77%. Table 7 provides the class-based performance results of Xception model.

Table 7 displays the following parameters: recall, precision, F1-score, accuracy, and loss. Loss values and accuracy show that the Xception model is performing well. This model has a high accuracy value and a low loss value. The result indicates the effectiveness of the methodology. Fig. 10 provides an overview of the outcomes determined using the data and the epochs.

This model works very well for both the training and testing dataset. After 25 epochs, both the training and testing curve were much constant. As a result, both curves provide quite the same result in maximum time.

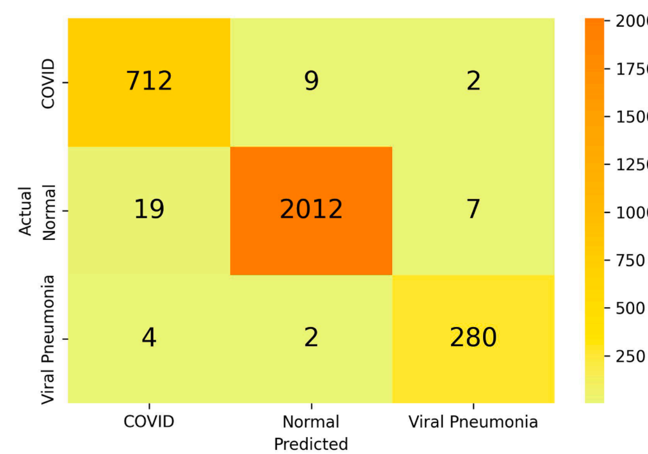

Fig. 9. Proposed Densenet-169 model confusion matrix.

Table 7
The statistical results of class based performance of proposed Xception model.

Class	Task	Precision	Recall	F1-score	Loss	Accuracy
	Covid-19	0.96	0.98	0.97		
Three class	Pneumonia	0.97	0.96	0.97	0.089	0.970
	Normal	0.99	0.96	0.97		
Two Class	Covid-19	0.99	0.95	0.97		
	Normal	0.96	0.99	0.98	0.081	0.973

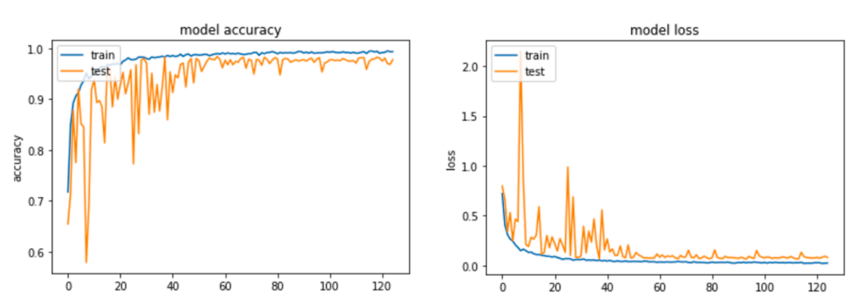

Fig. 8. Model accuracy and model loss curve of the proposed DenseNet-169 model.

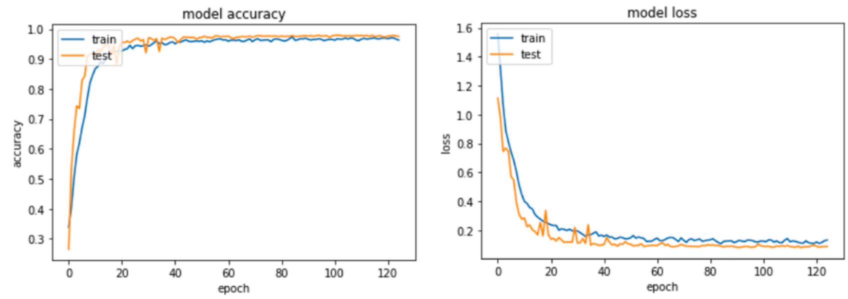

Fig. 10. Model accuracy and model loss curve of proposed Xception model.

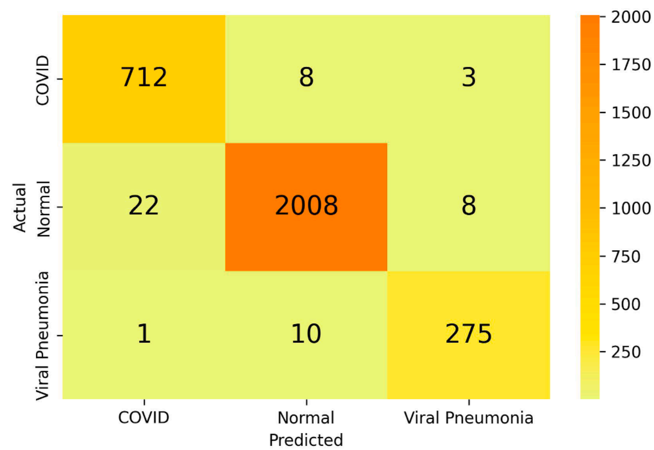

Fig. 11. Proposed Xception model confusion matrix.

Table 8
The statistical results of class based performance of proposed VGG-16.

Class	Task	Precision	Recall	F1-score	Loss	Accuracy
Three class	Covid-19	0.98	0.98	0.98	0.076	0.970
	Pneumonia	0.98	0.98	0.98		
	Normal	0.98	0.97	0.97		
Two Class	Covid-19	0.99	0.95	0.96	0.090	0.975
	Normal	0.95	0.99	0.97		

Furthermore, there were very few ups and down in the curve, which shows it is a good model. The confusion matrix of Xception algorithm with the provided dataset is given below in Fig. 11:

4.1.3. Performance of VGG-16 Algorithm

Using chest radiograph pictures, VGG-19 was trained to detect both two and three-class. The dataset was split into training and testing sets, with 20% of the data used for model testing and 80% utilized for model training. A validation accuracy of 98.40% was attained by the model by epoch 125.

Table 8 displays precision, recall, F1-score, loss values and accuracy of the model after training and testing it. The results demonstrate the effectiveness of the suggested methodology in identifying COVID-19. Fig. 12 provides an overview of the outcomes determined using the data and the epochs.

In the model accuracy and loss curve, at first, the model doesn't fit well. There were some ups and downs in the curve. After 35 epochs, the models show a good result. The curve is not much constant as the Xception model, although it provides good outcomes with both the training and testing dataset. The confusion matrix of VGG-16 algorithm with the provided dataset is given below in Fig. 13:

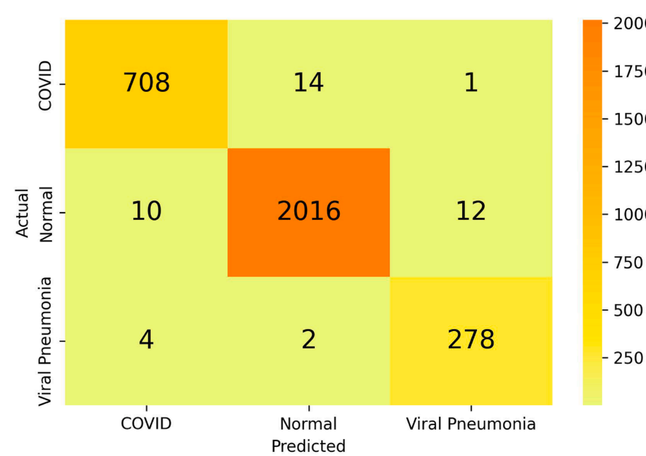

Fig. 13. Proposed VGG-16 model confusion matrix.

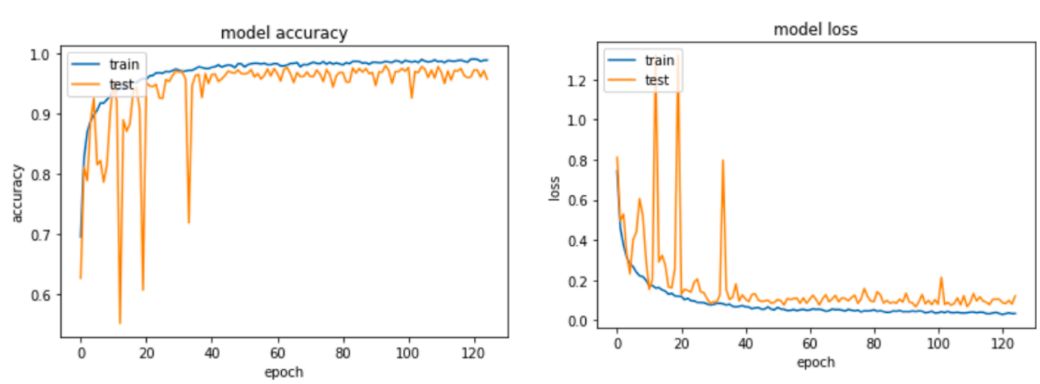

Fig. 12. Model accuracy and model loss curve of proposed VGG-16 model.

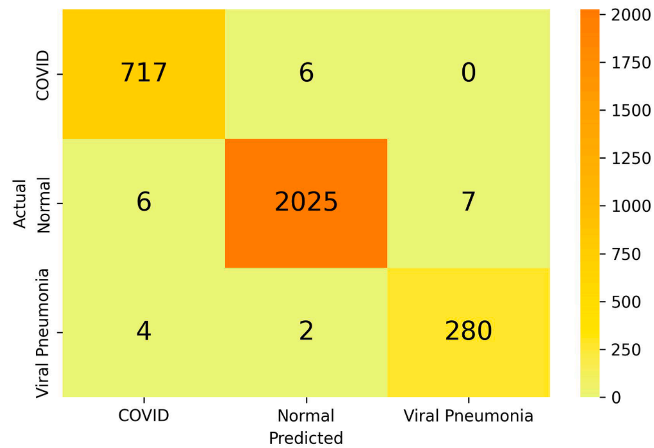

Fig. 14. Proposed stacking ensemble model confusion matrix.

4.1.4. Stacking Ensemble model

The proposed model is trained and tested using COVID-19 radiography datasets . We formed a stacking ensemble model that includes Xception, Densenet-161, and VGG-16. The proposed model's performance is compared against baseline classifiers: Densenet-161, Xception, and VGG-16. These models are represented by a softmax layer and a fully linked layer. As a result, the suggested model scored 0.98 for the F1 score and 0.986 for the accuracy scale. In the proposed stacking ensemble model, true reports are more and false reports are less than the other three models mentioned earlier. The confusion matrix of stacking ensemble model with the provided dataset is given below in Fig. 14:

Fig. 15 compares the proposed stacking ensemble model's performance with those of existing models for the mentioned dataset in terms of precision, recall, accuracy, and F1-score for both binary classification and multi-class classification. Table 9 below shows the results in details.

Comparison of accuracy of Stacking ensemble model with other three modified transfer learning model is given in Fig. 15:

4.1.5. Comparison with Related Work Methods

Table 10 shows a comparison of several existing models with the proposed methodology where we used the same dataset.

In Fig. 16 shows that the accuracy of the proposed methodology is better than the others existing models described in the related works section.

The suggested model exhibits superior results when compared to the current literature. This has a higher accuracy, F1-score, recall, and precision value than the other pre-trained models. Modified DenseNet-169, Xception, and VGG-16 separately predict better; it got an accuracy of 97.4%, 97%, and 97.5% in the case of multiclass classification and 98.1%. 97.3%, 97.1% respectively. After changing the weights of each model and stacking the ensemble of these three models, the accuracy increased to 99.10% for binary and 98.60% for multi-classification, which is better than the other related models.

Table 9
The statistical results of class based performance of stacking ensemble algorithm.

Class	Task	Precision	Recall	F1-score	Accuracy
Three class	Covid-19	0.98	0.99	0.99	0.986
	Pneumonia	0.99	0.98	0.98	
	Normal	0.98	0.98	0.98	
Two Class	Covid-19	0.99	0.97	0.98	0.991
	Normal	0.98	1.00	0.99	

Table 10
Comparison proposed methodology with other models.

Author's	Number of class	Precision	Recall	f1-score	Accuracy (%)
Lafraxo et al. [21]	2	0.9369	0.9366	0.9865	98.62
	3	0.9369	0.9366	-	95.77
Ahmed et al. [22]	3	0.92	0.89	0.898	90.64
Rahman et al. [23]	3	0.9455	0.9456	0.9453	95.11
Loey et al. [24]	3	0.9608	0.9604	0.9605	96.00
Aggarwal et al. [25]	3	0.9766	0.9733	0.9733	97.00
Lasker et al. [26]	3	0.9512	0.9489	0.95	96.50
Sahin et al. [27]	2	-	-	0.97	96.71
Ramadhan et al. [28]	2	0.96	0.9085	0.95	97
	3	-	0.9360	-	95.14
Proposed Method	2	0.97	0.96	0.97	99.10
	3	0.98	0.98	0.98	98.60

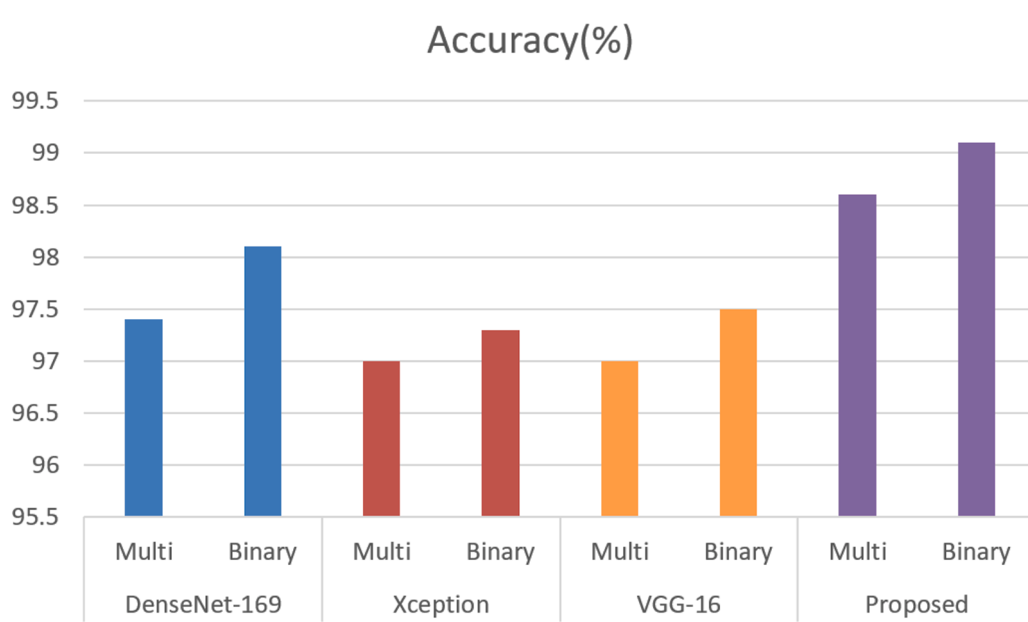

Fig. 15. Comparison of other pre-trained models with proposed stacking ensemble model.

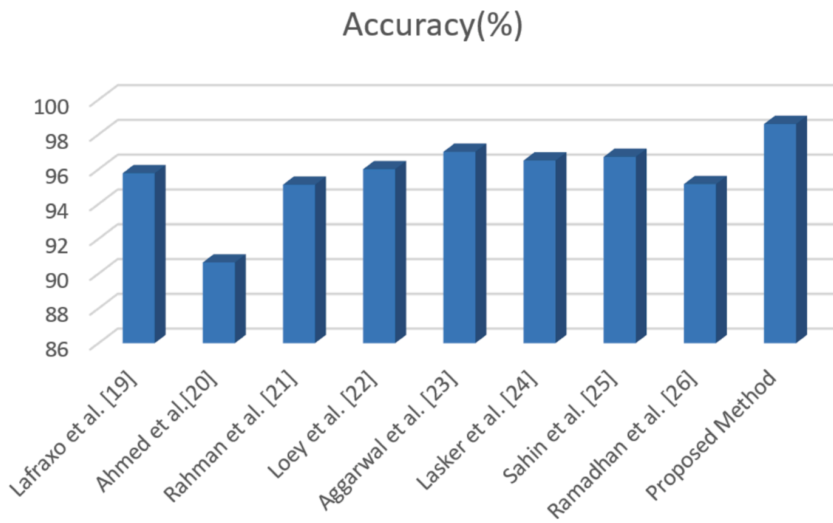

Fig. 16. Comparison of proposed model with other models.

Table 11
Used tools for the interface.

	Name	Version
Blockchain and solidity	Node.js	16.13.2
	Solidity	5.0.5
	Remix IDE	0.8.17
	Metamask	10.8
	Web3	1.8.1

4.1.6. *Implementation of Decentralized Application*

A decentralized application is developed for storing confidential data securely. To build the blockchain app, first, we have to make sure the environment is ready. For that, we have to install Node.js, install a local Ethereum network called Ganache, install a development framework called Truffle, and add the Metamask Extension to the web browser. Users can manage their Ethereum-based digital assets and engage with decentralized apps (DApps) on the Ethereum network using MetaMask, a well-known cryptocurrency wallet and browser extension. In order to access the decentralized world of cryptocurrency and DApps, MetaMask simplifies the procedure by giving consumers a nice user experience. After that, for developing the front-end, we used HTML, CSS, JS, Web3js, and React, and for building smart contracts, we used Solidity Language and the tools: Visual Studio Code and Remix. We have written web3js code in the front-end (UI) to establish connections and interactions between the UI and the smart contract through Metamask. After that, we deployed Smart Contract on Ganache's local P2P Ethereum network. Ganache is used for the building and testing of blockchain applications based on Ethereum. It enables developers to install decentralized apps (DApps), test smart contracts, and simulate various network situations in a local blockchain environment. The tools used to build this application are provided in Table 11 :

After image detection, the images need to be stored in IPFS for ensuring security and privacy. A decentralized application is built so that only authorized persons can make transactions.

Fig. 17 shows that only authorized persons can store patients' sensitive data. After submitting patients' sensitive data and images, the smart contract sends the image to IPFS, a hash value is created for every image in IPFS, and return the hash value to the smart contract; then the smart contract stores the hash value of the image which is sent by the IPFS and also stores patient's sensitive data in the blockchain. For cross-checking whether the image is saved in the IPFS or not, this URL has to be used https://ipfs.io/ipfs/hashvalue and is shown in Fig. 18.

The smart contract is deployed into a decentralized platform like

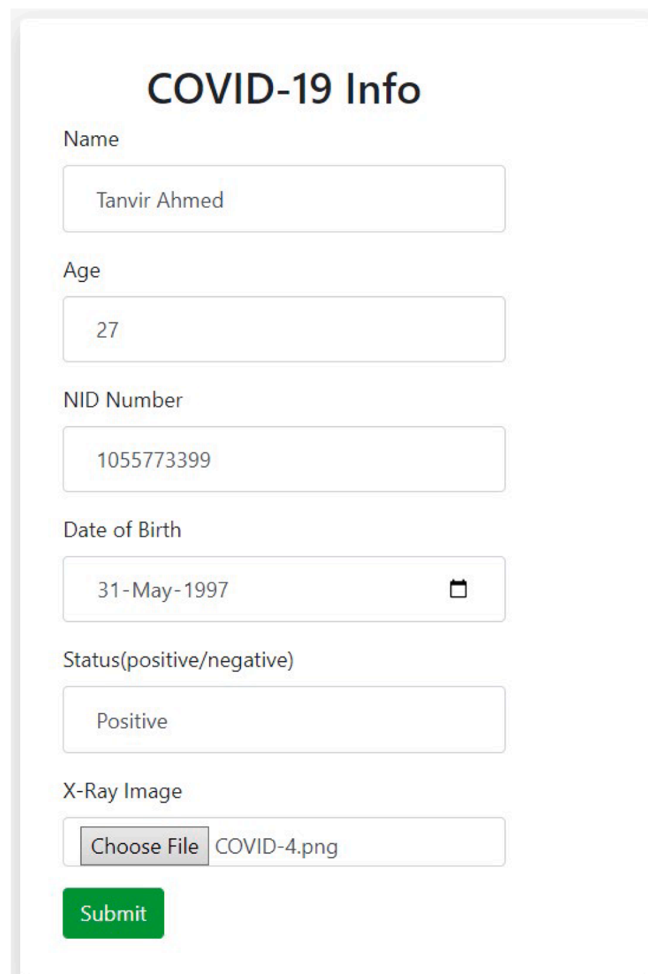

Fig. 17. Input patients' confidential data.

blockchain. Doing transactions in the smart contract requires a crypto wallet called metamsk. The metamask allows the smart contract to store files in the blockchain. Metamask allows a transaction when it has enough ETH or BTC in its own account. After submitting the image, a transaction confirmation notification is sent to the authorized user

Fig. 18. Saved image in IPFS.

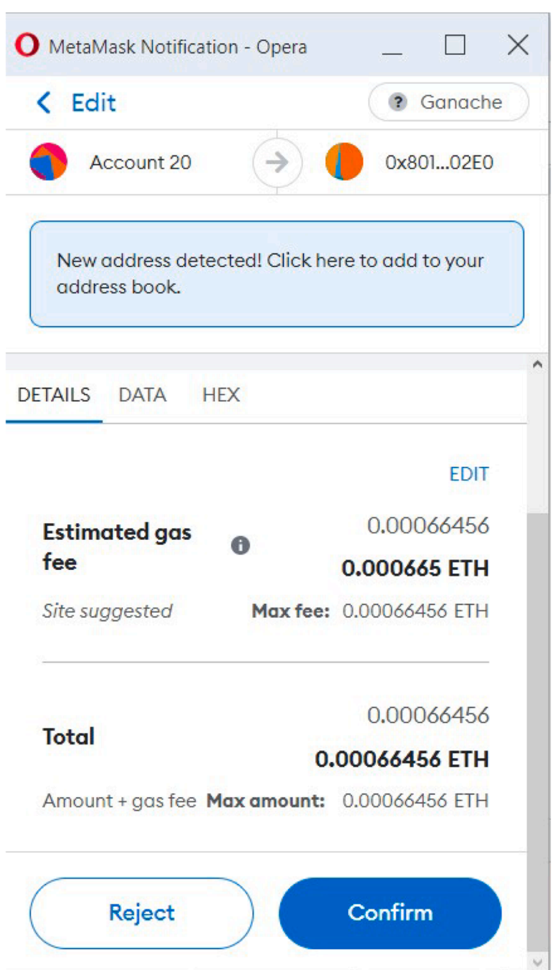

Fig. 19. Metamask transaction notification.

through metamask, which is shown in Fig. 19. Through the decentralized application, any user can check the report of any patient using national ID number. In the local database, a hash is saved with corresponding to a national ID number, so any user can easily check the report which is shown in Fig. 20.

After every successful transaction, a block is created in Ethereum blockchain. Fig. 21 shows the transaction details. After adding some patients' confidential data by smart contract, ETH has been reduced for each transaction. From metamask wallet, the balance can be checked to determine how much ETH or BTC is remaining.

4.2. Discussion

The results demonstrate that the suggested stacking ensemble-based deep learning model can automatically recognise and retrieve crucial information from CXRs, that is relevant to the diagnosis. Many investigations have suggested DL techniques in the literature to construct a trustworthy diagnosis model. In our research, we tried to find the best model for automatically detecting COVID-19 by testing different models on a large dataset and considering various factors.

The three models are trained to classify COVID-19 cases in the categorization mentioned above. The models DenseNet-169, Xception, and VGG-16 are examined for classification tasks and then combined them using stacking ensemble method. The suggested stacking ensemble model achieves test accuracy of 99.10% and an F1-score of 0.97 for binary class and 98.60% and 0.98 for multiclass, which indicates its usefulness in the diagnosis of COVID-19. The result analysis shows a higher accuracy rate compared to other state-of-the-art works. Additionally, the operational costs of X-ray equipment are comparatively modest since they require less reagent maintenance than RT-PCR.

In future research, our aim is to provide training for additional contemporary DL approaches, including hybrid methods and well-known techniques that are widely used. For future use, more chest image datasets that are publicly available will be gathered. Accuracy can be increased by utilizing the suggested model architecture if more chest X-ray samples can be used for training the model. High-quality data is required to improve the performance of deep learning models. Creating and annotating data, as well as metadata, can improve the performance of the deep learning model. The deep learning approach will be beneficial for future large-scale investigations aimed at finding "invisible cases," such as estimating the number of individuals who may be sick but do not exhibit symptoms.

5. Conclusion

To stop COVID-19 infection from spreading to other people, identifying COVID-19 is crucial as early as possible. In this article, we developed a stacking ensemble classification strategy to quickly and accurately diagnose COVID-19, utilizing chest radiograph images. The DenseNet-169, Xception, and VGG-16 models are stacked to create the proposed model. The suggested stacking ensemble model outperformed baseline models and other existing models.

The proposed stacking ensemble model showed an accuracy of 98.60% for COVID-19 identification. Our framework ensures patient's privacy using smart contracts on blockchain networks. The COVID-19 pandemic's vulnerable situation has highlighted the need for a single-source Blockchain-based pandemic health record management system to address the various present and future challenges. The most critical issue to solve the problems mentioned above is to use blockchain technology to store, share, and access data as the sole provider of information (database). According to future plans, the studies will include radiological images, such as CT and MRI to train the stacking ensemble model.

In this work, we include three classifications to build the stacking ensemble. We aim to investigate more pre-trained classifiers in future.

Declaration of Competing Interest

The authors declare that they have no known competing financial

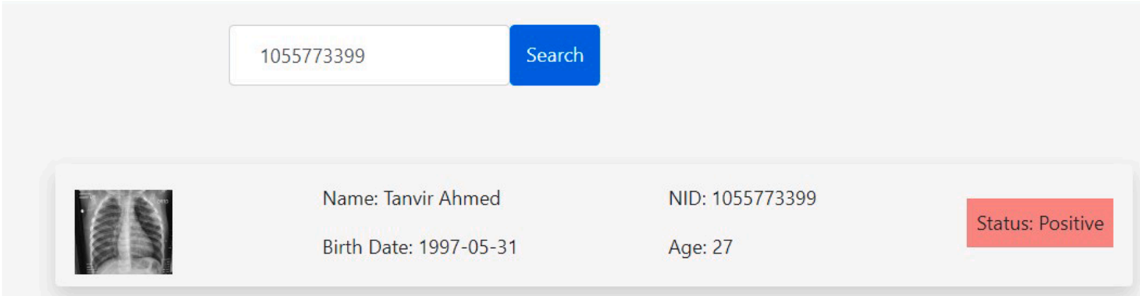

Fig. 20. Searching report using ID.

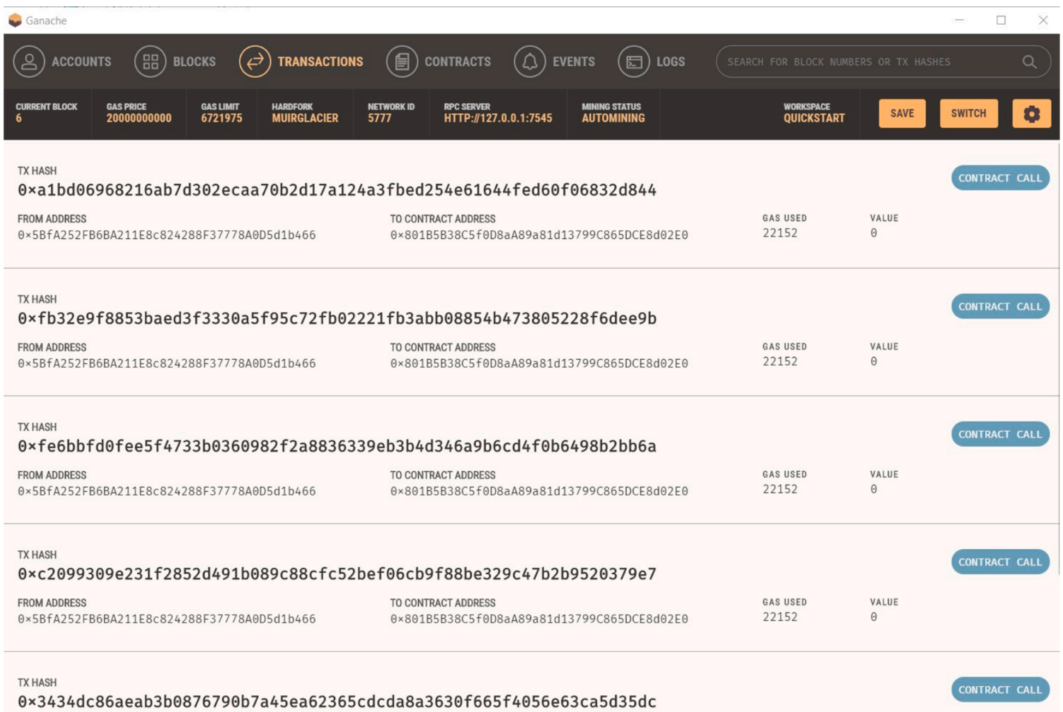

Fig. 21. Transactions shown in Ganache.

interests or personal relationships that could have appeared to influence the work reported in this paper.

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